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What is Azure?

- Cloud computing platform to build, deploy and manage applications and services
- Provides infrastructure through Microsoft-managed data centers
- Supports hosting web services, running virtualized computers, and custom software solutions

What is Cloud Computing?

Definition:

- Delivery of computing services via the internet.
- Services cover like virtual machines, storage, databases, and networking.

Key Aspects:

- Internet-based delivery offers flexibility and accessibility.
- Doesn't rely on physical infrastructure like traditional data centers.
- Rapid scalability without the need for constructing new physical setups.
- Instantly expands IT infrastructure using the cloud so avoids delays.

What is Shared Responsibility Model?

Traditional Datacenter Responsibilities:

- Company maintains physical space, security, and server maintenance.

Shift to Shared Responsibility Model

- Cloud provider and consumer share responsibilities.

Varied Responsibilities in the Cloud:

- Responsibilities differ based on cloud service types: IaaS, PaaS, SaaS.
- IaaS leans towards consumer responsibility, focusing on managing infrastructure.
- PaaS distributes responsibilities evenly between provider and consumer.
- SaaS places more responsibility on the cloud provider for service maintenance.

Key Areas of Responsibility:

- Always Consumer:
 - Data stored in the cloud.
 - Device access to the cloud (e.g., phones, computers).
 - Accounts and identities management within the organization.
- Always Cloud Provider:
 - Physical datacenter, network, and hosts.

	Responsibility	SaaS	PaaS	IaaS	On-prem
Responsibility always retained by the customer	Information and data	Customer	Customer	Customer	Customer
	Devices (Mobile and PCs)	Customer	Customer	Customer	Customer
	Accounts and identities	Customer	Customer	Customer	Customer
Responsibility varies by type	Identity and directory infrastructure	Shared	Shared	Customer	Customer
	Applications	Microsoft	Shared	Customer	Customer
	Network controls	Microsoft	Shared	Customer	Customer
	Operating system	Microsoft	Microsoft	Customer	Customer
Responsibility transfers to cloud provider	Physical hosts	Microsoft	Microsoft	Microsoft	Customer
	Physical network	Microsoft	Microsoft	Microsoft	Customer
	Physical datacenter	Microsoft	Microsoft	Microsoft	Customer

 Microsoft
  Customer
  Shared

What are Cloud Models?

Private Cloud:

- Used by a single entity, offering greater control.
- Hosted on-site or in a dedicated offsite datacenter, potentially managed by a third party.
- Comes with higher costs and fewer benefits compared to public cloud.

Public Cloud:

- Built, controlled, and maintained by third-party cloud providers.
- Accessible to anyone seeking cloud services, offering general public availability.

Hybrid Cloud:

- Utilizes both public and private clouds in an interconnected environment.
- Offers flexibility and security choices by selectively using public and private cloud infrastructures.

Multi-cloud:

- Uses multiple public cloud providers for various reasons, such as different features or migration needs.
- Manages resources and security across multiple public cloud environments.

Azure Arc:

Technology aiding in managing diverse cloud environments, including public, private, hybrid, and multi-cloud setups.

Azure VMware Solution:

Enables running VMware workloads in Azure, allowing seamless integration and scalability for migrations from private to public or hybrid clouds.

Public cloud	Private cloud	Hybrid cloud
No capital expenditures to scale up	Organizations have complete control over resources and security	Provides the most flexibility
Applications can be quickly provisioned and deprovisioned	Data is not collocated with other organizations' data	Organizations determine where to run their applications
Organizations pay only for what they use	Hardware must be purchased for startup and maintenance	Organizations control security, compliance, or legal requirements
Organizations don't have complete control over resources and security	Organizations are responsible for hardware maintenance and updates	

What are Consumption Based Models?

- **IT Infrastructure Models:** CapEx (Capital Expenditure) and OpEx (Operational Expenditure).
- **CapEx vs. OpEx:** CapEx involves one-time purchases (buildings, data centers, vehicles), while OpEx covers ongoing expenses (renting, leasing, cloud services).
- **Cloud Computing and OpEx:** Cloud computing operates on a consumption-based OpEx model.
- **How Cloud Computing Works:** Payment for IT resources used, not for physical infrastructure or maintenance costs.

Benefits of Consumption-based Model:

- No upfront costs.
- No need to manage costly infrastructure that might be underutilized.
- Flexibility to scale resources up or down as needed.

Comparison with Traditional Datacenter:

- Estimation in traditional datacenters: Overestimation leads to wastage, underestimation affects performance.
- Cloud-based model removes the need for precise estimation.

Cloud Pricing Models:

- Pay-as-you-go model for cloud computing.
- Benefits include cost planning, efficient infrastructure management, and scalability.

High Availability and Scalability in the Cloud

When building or deploying a cloud application, two crucial considerations are uptime (or availability) and the ability to handle demand (or scale).

High Availability:

- **Focus:** Ensures resources are available when needed during application deployment.
- **Objective:** Maximum availability despite disruptions or events.

Scalability:

- **Definition:** Ability to adjust resources to meet varying demand, a key cloud computing benefit.
- **Handling Peak Traffic:** Allows adding resources to cope with sudden increases in demand, preventing system overload.
- **Cost Efficiency:** Avoids overpayment by operating on a consumption-based model—pay only for what you use.

Scaling Types:

- **Vertical Scaling:** Adjusts resource capabilities, e.g., adding or reducing CPUs and RAM.
- **Horizontal Scaling:** Modifies the number of resources, e.g., adding or removing virtual machines or containers.

Reliability and Predictability in the Cloud

Reliability:

- **Definition:** System's ability to recover from failures and maintain functionality.
- **Automatic Resilience:** Applications can shift regions automatically for increased reliability.

Predictability:

- **Confident Development:** Predictability fosters development confidence.

Performance Predictability:

- **Resource Predictions:** Predicting resources for a positive user experience.
- **Supporting Concepts:** Autoscaling, load balancing, and high availability aid in predictability.
- **Autoscaling:** Dynamically adjusts resources based on demand.
- **Load Balancing:** Ensures even resource distribution for optimal performance.

Cost Predictability:

- **Cost Forecasting:** Predicting and forecasting cloud spending.
- **Real-Time Monitoring:** Tracking resource use in real-time for efficient utilization.
- **Data Analytics:** Identifying patterns for better resource deployment planning.
- **Tools for Estimation:** Total Cost of Ownership (TCO) and Pricing Calculator provide spending estimates.

Benefits of Security and Governance in the Cloud

Governance:

- **Templates:** Ensure deployed resources meet corporate standards and regulatory requirements.
- **Automated Patching:** Depending on the model, patches and updates can be automatically applied.

Security:

- **Tailored Solutions:** Cloud offers security solutions matching specific needs.
- **Control Options:** Infrastructure as a service provides maximum control over security. Platform or software as a service automates patches and maintenance.
- **Resilience to Attacks:** Cloud providers adept at handling DDoS attacks enhance network robustness and security.

Benefits of Manageability in the Cloud

- **Automated Scaling:** Resources scale dynamically based on demand.
- **Automated Health Monitoring:** Monitors resource health and replaces failing resources automatically.
- **Real-time Alerts:** Automatic alerts based on configured metrics provide real-time performance awareness.

Management in the Cloud are through:

- Web Portal
- Command Line Interface
- APIs
- PowerShell

Infrastructure as a Service (IaaS)

Definition: IaaS is a flexible cloud service where the provider manages hardware, connectivity, and security, while users control OS, network, database, and storage. It provides you the maximum amount of control for your cloud resources.

Responsibilities:

- **Cloud Provider:** Handles physical infrastructure and internet access.
- **User:** Manages OS, network, database, storage, and security.

Scenarios:

- **Lift-and-Shift Migration:** Move on-premises workloads to IaaS.
- **Testing and Development:** Quickly replicate and manage environments.

Platform as a Service (PaaS)

Definition: PaaS is a cloud service handling infrastructure, OS, databases, and development tools. Users focus on networking and security.

Responsibilities:

- **Cloud Provider:** Manages extensive services including infrastructure, OS, databases, and development tools.
- **User:** Concerned with networking, security, and application specifics.

Shared Responsibility Model: Similar to IaaS, but with the provider managing a broader range of services.

Scenarios:

- **Development Framework:** Ideal for building cloud-based applications with built-in components, reducing coding efforts.
- **Analytics or Business Intelligence:** Offers tools for data analysis and prediction, enhancing decision-making.

Advantages: Provides a complete development environment without infrastructure maintenance hassles.

Example Use Case: Suited for developers building cloud applications and organizations leveraging analytics without managing underlying infrastructure.

Software as a Service (SaaS)

Definition: SaaS is a comprehensive cloud service providing fully developed applications on a rental basis. Examples include email, financial software, and messaging applications.

Flexibility: While the least flexible, SaaS is the easiest to deploy, requiring minimal technical knowledge.

Shared Responsibility Model:

- **User:** Responsible for data input, device connections, and user access.
- **Cloud Provider:** Manages physical security, power, network connectivity, and application development and patching.

Scenarios:

- **Email and Messaging:** Common applications delivered through SaaS.
- **Business Productivity:** SaaS for applications enhancing business operations.
- **Finance and Expense Tracking:** Utilizing SaaS for financial management.

Advantages: Quick setup with minimal technical expertise; extensive responsibility on the cloud provider.

Example Use Case: Ideal for businesses seeking ready-to-use applications like email, messaging, and productivity tools without the need for in-depth technical

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Azure Physical Infrastructure

Datacenters:

- Large facilities like corporate datacenters.
- Equipped with racks, dedicated power, cooling, and networking.

Regions:

- Geographical areas with at least one nearby datacenter.
- Azure balances resources intelligently within each region.

Availability Zones:

- Physically separate datacenters within a region.
- Ensure high availability; if one zone fails, others continue

Region Pairs:

- Paired regions for redundancy, e.g., West US with East US.
- Prioritized restoration in case of extensive outages (If Azure has a big issue, they fix one region in each pair fast, so your apps always have a backup).
- Planned updates rolled out one region at a time for minimal downtime (When Azure updates, they do it one region at a time in pairs, keeping your apps running smoothly).
- Data stays in the same geography as its pair, except for Brazil South (Your data stays in the same area as its backup (except Brazil South) for legal reasons, like taxes and laws)

Sovereign Regions:

- Isolated instances for compliance or legal reasons.
- Examples include US DoD Central, US Gov Virginia, China East, etc.

Azure Management Infrastructure

Resource: Basic building block (e.g., VMs, databases).

Resource Group: A grouping of resources.

- Resources are placed within resource groups.
- Actions on a resource group apply to all resources within it.
- No nesting of resource groups allowed.

Azure Subscriptions:

- Necessary for accessing Azure services.
- Linked to an Azure account.

Two subscription types:

- **Billing Boundary:**

- Determines billing for Azure usage.
- Multiple subscriptions for diverse billing needs.
- Separate billing reports and invoices for each subscription.
- **Access Control Boundary:**
 - Manages access through policies.
 - Allows creating distinct subscriptions for different organizational structures.
 - Access policies applied at the subscription level.

Azure Management Groups:

- Efficiently manages access, policies, and compliance.
- Organizes subscriptions into containers called management groups.
- Governance conditions applied to management groups.
- Hierarchy:
Management groups > Subscriptions > Resource groups > Resources.
- Allows for a flexible structure for unified policy and access management.

Benefits:

- Scales for large organizational setups.
- Settings applied at the management group level inherit to all its subscriptions and resources.
- Management groups can be nested for intricate management needs.
- Inherited by all sub-management groups, subscriptions, and resources.
- Supports up to 10,000 management groups in a single directory.
- A management group tree supports up to six levels of depth.
- Each management group and subscription can have only one parent.
- Enhances clarity in the hierarchy.

Azure Virtual Machines

- A virtual machine is essentially a software-based emulation of a physical computer, running an operating system and applications.
- Azure Virtual Machines (VMs) provide Infrastructure as a Service (IaaS) in the cloud.
- Offers flexibility, customization, and control over the operating system and software.

VM Image:

- VMs can be created from custom images or preconfigured templates.
- Images serve as templates, including the OS and software, facilitating rapid VM provisioning

Scale Sets:

- Manage a group of identical, load-balanced VMs efficiently.
- Automates configuration, updates, and load balancing.
- Enables scalability based on demand or defined schedules.

Availability Sets:

Imagine you have a bunch of virtual machines (VMs) in the cloud, and you want to make sure they are always available and don't all go offline at once. This is where availability sets come in, acting like a safety net for your VMs.

Update Domain:

- VMs in the same update domain can be updated (rebooted) at the same time.
- This way, if one group is getting updated, the others are still up and running.
- Gives a 30-minute break between each group's update.

Fault Domain:

- VMs in different fault domains are in different houses (connected to different power and internet).
- If something goes wrong in one house (power or internet issue), VMs in other houses are not affected.

Benefits:

- Ensures not all VMs are updated or affected by issues simultaneously.
- Guards against problems like power failures or network issues.
- Keeps your applications running even during updates or unexpected hiccups.

Cost:

- No extra cost for setting up availability sets.
- You only pay for the virtual machines you create.

Create Azure Virtual Machine with Nginx Installation

Azure CLI (Command-Line Interface)

- Efficient for automating tasks.
- Allows scripting and quick execution of commands.
- Example: `az vm create` for VM creation.

PowerShell:

PowerShell is a task automation framework and scripting language designed for system administrators. It provides a command-line shell and a scripting language to automate repetitive tasks, configuration management, and streamline system administration workflows. PowerShell commands, called cmdlets, facilitate the interaction with various components of the Windows operating system and other applications.

Azure Resource Manager (ARM)

Azure Resource Manager (ARM) is the deployment and management service provided by Microsoft Azure. It allows users to define the resources they need in a declarative template, known as an ARM template. These templates enable consistent and repeatable deployment of Azure resources, ensuring that the infrastructure is provisioned and configured in a predictable manner. ARM simplifies resource lifecycle management, including creation, update, and deletion, providing a unified control plane for Azure resources.

Cloud Shell:

Azure Cloud Shell is a cloud-based shell provided by Microsoft Azure, accessible through the Azure portal. It offers a command-line environment with common Azure tools preinstalled and configured, eliminating the need for users to install and maintain these tools locally. Cloud Shell allows seamless

management of Azure resources from anywhere with internet access, providing a consistent and secure environment for executing commands, scripts, and managing cloud infrastructure.

Nginx:

- Popular web server and reverse proxy.
- Known for its speed and scalability.
- Used to serve web content efficiently.
- Frequently employed in web hosting environments.

1. From Cloud Shell, run the following `az vm create` command to create a Linux VM:

```
Azure CLI Copy

az vm create \
  --resource-group "learn-cd68636b-de69-4f24-84d7-520a3c087a13" \
  --name my-vm \
  --public-ip-sku Standard \
  --image Ubuntu2204 \
  --admin-username azureuser \
  --generate-ssh-keys
```

2. Run the following `az vm extension set` command to configure Nginx on your VM:

```
Azure CLI Copy

az vm extension set \
  --resource-group "learn-cd68636b-de69-4f24-84d7-520a3c087a13" \
  --vm-name my-vm \
  --name customScript \
  --publisher Microsoft.Azure.Extensions \
  --version 2.1 \
  --settings '{"fileUri":["https://raw.githubusercontent.com/MicrosoftDocs/msdntest/master/scripts/ConfigureNginx.ps1"]}' \
  --protected-settings '{"commandToExecute": "dotnet ./ConfigureNginx.ps1"}'
```

Azure Virtual Desktop

Overview:

- Azure Virtual Desktop is a cloud-based desktop and application virtualization service.
- It allows access to a cloud-hosted version of Windows from any location.
- Works seamlessly across devices and operating systems, accessible through remote desktops or modern browsers.

Security Features:

- Centralized security management using Microsoft Azure AD.
- Enables multifactor authentication for enhanced user sign-in security.
- Granular role-based access controls (RBACs) to secure data access.

Enhanced Security:

- Desktops and apps run in the cloud, reducing the risk of confidential data on local devices.
- User sessions are isolated in both single and multi-session environments.

Multi-Session Deployment:

- Supports Windows 10 or Windows 11 Enterprise multi-session.
- Enables multiple concurrent users on a single virtual machine (VM).

Containers

- Containers are a virtualization environment, allowing multiple instances of an application on a single host machine.
- Unlike virtual machines, containers are lightweight, agile, and don't require managing the operating system.
- They enable dynamic creation, scaling, and quick restarts, providing flexibility and responsiveness to changes in demand.
- Virtual machines are limited to a single operating system per instance, while containers share the host OS kernel.
- Containers offer a lighter weight and more agile method, allowing faster response to changes and interruptions.
- Docker is a popular container engine, and Azure supports Docker for containerized applications.

Azure Container Services

Azure Container Instances (ACI):

- Fastest and simplest way to run a container in Azure.
- Platform as a Service (PaaS) offering, eliminating the need to manage virtual machines.

Azure Container Apps:

- Similar to Container Instances but with added benefits like load balancing and scaling.
- Provides a Platform as a Service (PaaS) solution for containerized applications

Azure Kubernetes Service (AKS):

- Container orchestration service managing the lifecycle of containers.
- Simplifies fleet management when deploying and scaling containers efficiently.

Use Cases:

- Containers are often used in a microservice architecture, breaking solutions into smaller, independent pieces.
- Microservices allow for the separate maintenance, scaling, or updating of different components, enhancing flexibility.
- Example: A website split into containers for front end, back end, and storage, enabling independent scaling and maintenance.

Benefits:

- Containers provide efficient resource utilization, scalability, and independence for different components in a solution.

- Azure Container Instances, Container Apps, and Azure Kubernetes Service offer varying levels of container management and orchestration based on specific needs.

CONTAINER	VM
A lightweight alternative to full machine virtualization that involves encapsulating an application with its own operating environment	An operating system (OS) or application environment that is installed on software, which imitates dedicated hardware
Lightweight	Heavyweight
Provides OS virtualization	Provides hardware-level virtualization
All share the host OS	Each runs in its own OS
Requires less memory space	Requires more memory space
Startup time is in milliseconds	Startup time is in minutes
Ex: LXC, LXD, CGManager, Docker	Ex: VMware vSphere, Virtual Box, Hyper – V
Reduces IP management resources, reduces and simplifies security updates, requires minimum code to transfer, and upload workloads	Provides all OS resources available to apps, established management and security tools
Provides process-level isolation, which is less secure	Fully isolated and are more secure

Azure Functions

Definition: Azure Functions is a serverless compute option in Azure, designed for event-driven scenarios without the need for maintaining virtual machines or containers.

Serverless Computing:

Operates in a serverless environment, triggered by events rather than continuously running resources.

Benefits:

- Ideal for focusing on code without managing infrastructure.
- Well-suited for event-driven tasks and automatic scaling based on demand.
- Cost-efficient with charges based on actual CPU time used.

State Management:

- Stateless (default): Functions act as if restarted with each event.
- Stateful (Durable Functions): Allows tracking prior activity for complex workflows.

Flexibility:

- Functions serve as a general compute platform, offering flexibility to deploy in serverless or non-serverless environments based on app requirements.

Application Hosting Options

Azure provides various options for hosting applications, allowing flexibility based on your needs. Two common choices are Virtual Machines (VMs) and containers. VMs offer maximum control and familiarity, while containers provide isolation and efficient management. Another robust option is Azure App Service, a platform that simplifies hosting, supporting web apps, background jobs, mobile back-ends, and APIs. Here are key points about Azure App Service:

Azure App Service:

Enables hosting web apps, background jobs, mobile back-ends, and RESTful APIs without managing infrastructure.

- **Languages:** Supports .NET, .NET Core, Java, Ruby, Node.js, PHP, Python, and offers both Windows and Linux environments.
- **Deployment:** Facilitates automated deployments from GitHub, Azure DevOps, or any Git repo for continuous deployment.
- **Flexibility:** Hosts various app service styles, including web apps, API apps, WebJobs, and mobile apps.

Types of App Services:

- **Web Apps:** Supports ASP.NET, ASP.NET Core, Java, Ruby, Node.js, PHP, or Python on Windows or Linux.
- **API Apps:** Allows building REST-based web APIs with Swagger support and Azure Marketplace integration.
- **WebJobs:** Runs programs or scripts in the context of web, API, or mobile apps, often for background tasks.
- **Mobile Apps:** Quickly builds back ends for iOS and Android apps with features like cloud-based SQL database storage, social authentication, push notifications, and custom back-end logic.

App Service Features:

- **Integrated Deployment:** Streamlines deployment and management within the platform.
- **Security:** Enables endpoint security and quick scaling for high traffic.
- **High Availability:** Utilizes built-in load balancing and traffic manager for increased availability.
- **Common Infrastructure:** All app styles share the same infrastructure, making it flexible and efficient.

Azure Virtual Networking

Isolation and Segmentation:

- Multiple isolated virtual networks within Azure.
- Defined private IP address space for each virtual network.
- IP address range is not internet-routable.
- Divisible into subnets for further segmentation.
- Name resolution using Azure's built-in service or external DNS.

Internet Communications:

- Enable incoming internet connections by assigning public IP addresses.

- Resources accessible via public load balancers.

Communicate Between Azure Resources:

Virtual networks connect various Azure resources:

- VMs
- App Service Environment for Power Apps
- Azure Kubernetes Service
- Azure VM scale sets

Service endpoints facilitate connections to Azure resource types like SQL databases and storage accounts.

Communicate with On-Premises Resources:

Three connectivity mechanisms:

- Point-to-site VPN: External computer to corporate network.
- Site-to-site VPN: On-premises VPN device to Azure VPN gateway.
- Azure ExpressRoute: Dedicated private connectivity to Azure without internet travel.

Route Network Traffic:

- Default routing between subnets on connected virtual networks, on-premises networks, and the internet.
- Control over routing through:
 - Route tables defining traffic rules.
 - BGP for propagating on-premises BGP routes to Azure virtual networks.

Filter Network Traffic:

- Use Network Security Groups (NSGs) to define inbound and outbound security rules.
- Network Virtual Appliances (NVAs) act as specialized VMs for specific network functions like firewalls or WAN optimization.

Connect Virtual Networks:

- Virtual network peering for direct connection between two virtual networks.
- Private network traffic on Microsoft backbone network, avoiding the public internet.
- Global interconnected networks across separate regions.

User-Defined Routes (UDR):

- Control routing tables within a virtual network or between virtual networks.
- Greater control over network traffic flow.

Exercise - Configure Network Access

Task 1: Access Your Web Server

- Use **az vm list-ip-addresses** command to get VM's IP address. (Azure Command-Line Interface (CLI) command to fetch the IP address of your virtual machine)

- Run **curl** command to download the home page. (Command-line tool for making HTTP requests, used here to access the web server)
- Alternatively, attempt to access the web server from a browser.

Task 2: List Current Network Security Group Rules

- Execute **az network nsg list** to list associated NSGs. (Azure CLI command to list Network Security Groups (NSGs), which are used to control inbound and outbound traffic to network interfaces and VMs)
- Run **az network nsg rule list** to list rules associated with the NSG. (Azure CLI command to display rules configured in a Network Security Group)
- Use the command with **--query** and **--output** to display rule details in a readable table.

Task 3: Create Network Security Rule

- Run **az network nsg rule create** to create a rule named allow-http, allowing inbound access on port 80. (Azure CLI command to add a rule allowing HTTP traffic to your Network Security Group)
- Verify the configuration with **az network nsg rule list**.

Task 4: Access Your Web Server Again

- Retry the **curl** command to access the web server.
- After NSG update, check the web server response.

Azure Virtual Private Networks (VPNs)

VPN Basics:

- **Secret Tunnels in Networks:**
 - VPNs establish secure, encrypted pathways within networks.
- **Connect Trusted Networks:**
 - They connect two or more trusted private networks over an untrusted network, usually the public internet.
- **Encrypts Traffic for Safety:**
 - Traffic is encrypted during transit, preventing eavesdropping or unauthorized access.
- **Enable Secure Information Sharing:**
 - VPNs enable networks to safely and securely share sensitive information.

VPN Gateways:

- **Virtual Network Gatekeepers:**
 - VPN gateways are like virtual network gatekeepers managing and facilitating connections.
- **Enable Various Connections:**
 - On-premises to Virtual Networks (Site-to-Site):
 - Connects on-premises datacenters to virtual networks through a secure site-to-site connection.
 - Individual Devices to Virtual Networks (Point-to-Site):
 - Allows individual devices to securely connect to virtual networks.
 - Virtual Networks to Other Virtual Networks (Network-to-Network):
 - Links different Azure virtual networks securely.

- **Data Travels Safely in Private Tunnels:**
 - All data transfer is encrypted inside a private tunnel as it crosses the internet.

VPN Types:

Policy-Based:

- Static IP Routes Defined:
 - Specifies statically the IP address of packets that should be encrypted through each tunnel.

Route-Based:

- IPSec Tunnels as Virtual Interfaces:
 - Utilizes IPSec tunnels modeled as a network interface or virtual tunnel interface.

Use Route-Based VPN for:

- Connections Between Virtual Networks:
 - Connects multiple Azure virtual networks securely.
- Point-to-Site Connections:
 - Allows individual devices to establish secure connections.
- Multisite Connections:
 - Links to multiple sites securely.
- Coexistence with Azure ExpressRoute:
 - Works seamlessly with Azure ExpressRoute for a unified and secure network.

High-availability scenarios:

Ensures a robust and fault-tolerant VPN configuration.

High Availability (HA) Configurations:

Active/Standby:

- Backup Takes Over if Active Fails:
 - Even if there's planned or unplanned disruption, the standby instance automatically assumes responsibility.
- Minimal User Interruption:
 - Users may experience brief interruptions during failover, typically restored within seconds for planned maintenance and 90 seconds for unplanned disruptions.

Active/Active:

- Uses BGP Routing Protocol:
 - Allows load sharing and redundancy by assigning unique public IP addresses to each instance.
- Assigns Unique IPs to Instances:
 - Enhances high availability by deploying an additional VPN device on-premises.

ExpressRoute Failover:

- Secure Backup for ExpressRoute:
 - Configures VPN gateway as a secure failover path for Azure ExpressRoute connections, ensuring continuous connectivity.
- Ensures Continuous Connectivity:

- In high-availability scenarios, provides an alternative method of connectivity through the internet if there's a risk associated with an ExpressRoute circuit outage.

Zone-Redundant Gateways:

- Deployed in Availability Zones:
 - Offers resiliency, scalability, and higher availability in regions supporting availability zones.
- Enhances Resilience and Scalability:
 - Physically and logically separates gateways within a region, protecting on-premises network connectivity from zone-level failures.
- Requires Different SKUs:
 - Uses Standard public IP addresses instead of Basic public IP addresses for improved performance.

Azure ExpressRoute

What is Azure ExpressRoute?

Azure ExpressRoute is like a secure highway that allows you to extend your on-premises networks (your office, datacenter, etc.) into the Microsoft cloud (Azure and Microsoft 365) using a private connection facilitated by a connectivity provider. This dedicated connection is termed an ExpressRoute Circuit.

Key Features and Benefits:

- **Diverse Connectivity Options:**
 - Supports connectivity from various network types - any-to-any (IP VPN), point-to-point Ethernet, or virtual cross-connection through a connectivity provider at a colocation facility.
 - Doesn't rely on the public internet, ensuring more reliability, faster speeds, consistent latencies, and higher security.
- **Direct Access to Microsoft Cloud Services:**
 - Establishes connections to Microsoft cloud services, including Azure services, Microsoft Office 365, Microsoft Dynamics 365, Azure compute services (like Virtual Machines), and Azure cloud services (like Cosmos DB and Storage).
- **Global Reach:**
 - Enables global connectivity with ExpressRoute Global Reach, allowing data exchange across on-premises sites without relying on the public internet. For instance, connecting an office in Asia with a datacenter in Europe.
- **Dynamic Routing with BGP:**
 - Uses Border Gateway Protocol (BGP) for dynamic routing between your on-premises network and Microsoft cloud services.
- **Built-in Redundancy:**
 - Offers built-in redundancy at every peering location, enhancing reliability. You can configure multiple circuits for additional redundancy.

ExpressRoute Connectivity Models:

- **CloudExchange Colocation:**
 - Physical co-location of your facility at a cloud exchange (like an ISP), enabling a virtual cross-connect to the Microsoft cloud.
- **Point-to-Point Ethernet Connection:**
 - Establishes a direct point-to-point connection between your facility and the Microsoft cloud.
- **Any-to-Any Networks:**
 - Integrates your wide area network (WAN) with Azure, providing connections to your offices and datacenters.
- **Directly from ExpressRoute Sites:**
 - Connects directly into Microsoft's global network at strategically distributed peering locations. ExpressRoute Direct offers dual 100 Gbps or 10-Gbps connectivity, supporting Active/Active connectivity at scale.

Security Considerations:

- **Private Connection:**
 - Data doesn't travel over the public internet, reducing exposure to potential risks associated with internet communications.
- **Remaining Internet Traffic:**
 - Certain operations like DNS queries, certificate revocation list checking, and Azure Content Delivery Network requests may still use the public internet even with an ExpressRoute connection.

ExpressRoute provides a dedicated, secure, and reliable pathway for businesses to seamlessly integrate their on-premises infrastructure with the Microsoft cloud, ensuring efficient and secure data transfer.

Azure DNS

Azure DNS is like a digital phonebook for the internet, but hosted on Microsoft's cloud platform, Azure. It helps translate user-friendly domain names like "example.com" into computer-readable IP addresses like "192.0.2.1". Here's a breakdown of what Azure DNS offers:

- **Reliability and Performance:** Azure DNS is built on Microsoft's global network, ensuring that your domain names are always available and resolve quickly no matter where your users are located.
- **Security:** It comes with security features like role-based access control, activity logs, and resource locking to keep your DNS settings safe and secure.
- **Ease of Use:** You can manage your DNS records using the same tools and credentials as other Azure services, making it convenient and straightforward.
- **Customizable Virtual Networks:** Azure DNS allows you to create private DNS domains, giving you more control over your network configuration and allowing you to use custom domain names within your virtual networks.
- **Alias Records:** With alias record sets, you can easily link your domain to Azure resources like public IP addresses or Azure Traffic Manager profiles. These records automatically update if the underlying resource's IP address changes, ensuring seamless connectivity.

Remember, Azure DNS doesn't let you buy domain names directly, but once you've purchased a domain name from a domain registrar, you can manage its DNS settings using Azure DNS.

Storage Accounts

Definition:

- Unique namespace for Azure Storage data.
- Accessible globally over HTTP or HTTPS.
- Secure, highly available, durable, and scalable.

Types of Storage Accounts

Standard General-Purpose v2:

- Services: Blob Storage, Queue Storage, Table Storage, Azure Files.
- Redundancy Options: LRS, GRS, RA-GRS, ZRS, GZRS, RA-GZRS.
- Usage: Standard use for various scenarios, including blobs, file shares, queues, and tables.

Premium Block Blobs:

- Services: Blob Storage.
- Redundancy Options: LRS, ZRS.
- Usage: High transaction rates, smaller objects, low storage latency.

Premium File Shares:

- Services: Azure Files.
- Redundancy Options: LRS, ZRS.
- Usage: Enterprise or high-performance scale applications supporting SMB and NFS file shares.

Premium Page Blobs:

- Services: Page blobs only.
- Redundancy Options: LRS.
- Usage: Page blobs exclusively.

Storage Account Endpoints:

Naming Rules:

- Length: 3 to 24 characters.
- Characters: Numbers and lowercase letters only.
- Uniqueness: Must be unique within Azure.

Endpoint Formats:

- Blob Storage: <https://<storage-account-name>.blob.core.windows.net>
- Data Lake Storage Gen2: <https://<storage-account-name>.dfs.core.windows.net>
- Azure Files: <https://<storage-account-name>.file.core.windows.net>
- Queue Storage: <https://<storage-account-name>.queue.core.windows.net>
- Table Storage: <https://<storage-account-name>.table.core.windows.net>

Azure Storage Redundancy:

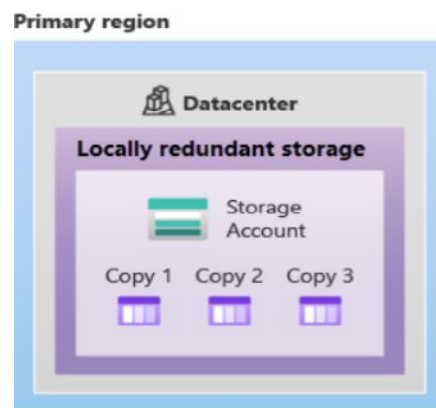
Purpose:

- Ensures data protection from planned and unplanned events.
- Guarantees availability and durability of storage account.

Primary Region Redundancy:

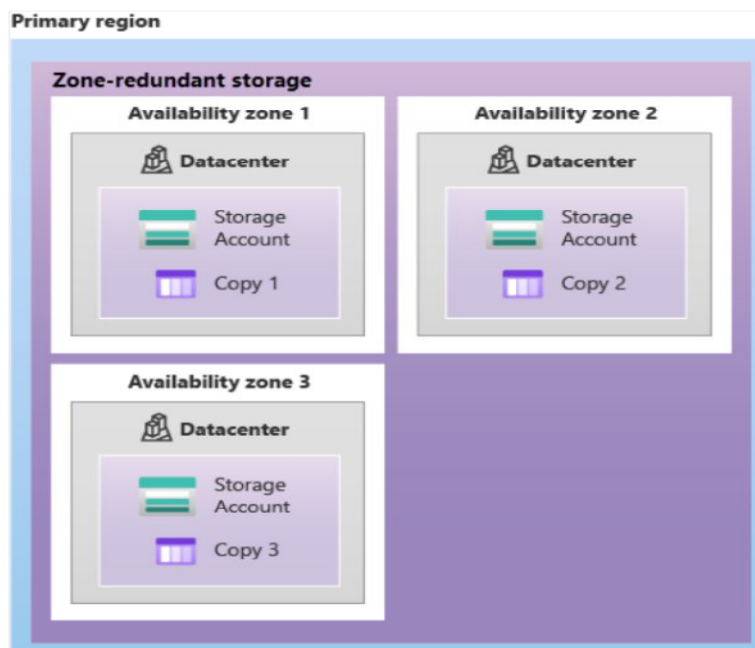
Locally Redundant Storage (LRS):

- Replicates data three times within a single data center.
- Provides 11 nines of durability (99.999999999%).
- Lowest-cost option but least durable.



Zone-Redundant Storage (ZRS):

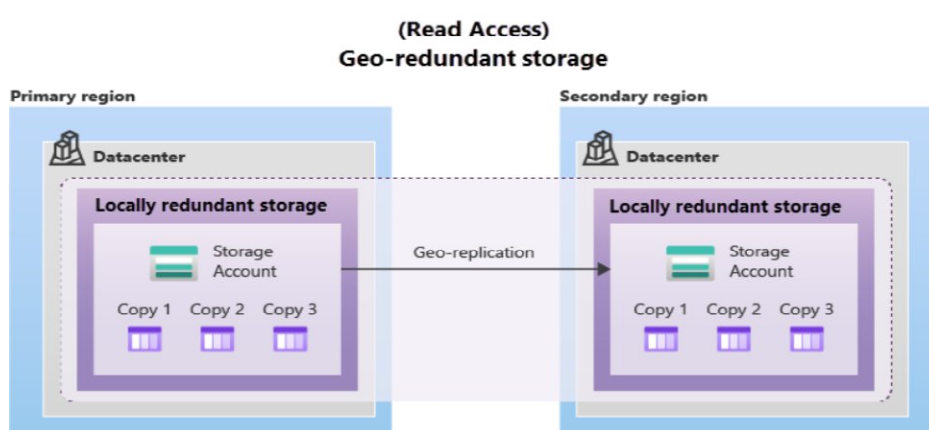
- Replicates data synchronously across three Azure availability zones.
- Offers at least 12 nines of durability.
- Recommended for high availability scenarios.



Secondary Region Redundancy:

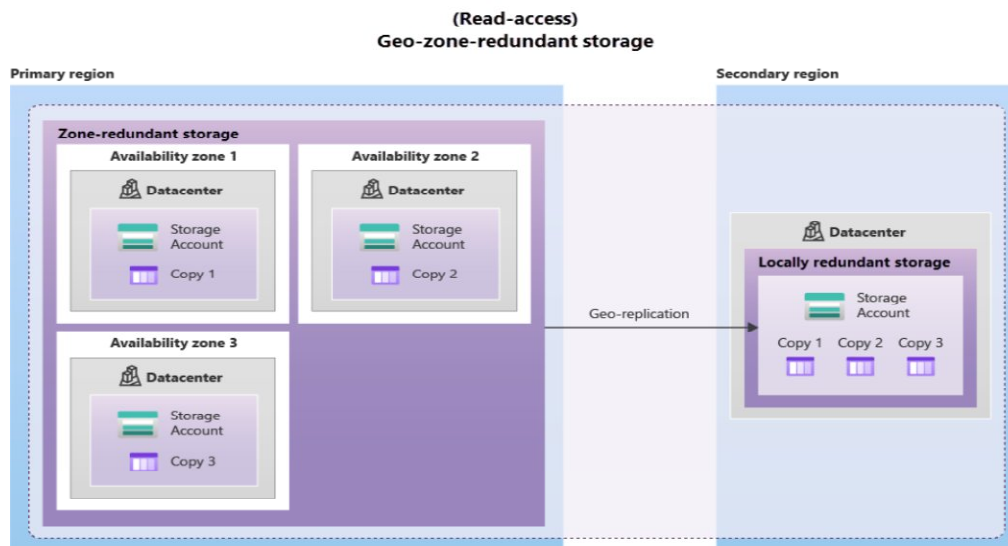
Geo-Redundant Storage (GRS):

- Copies data synchronously within primary region using LRS.
- Copies data asynchronously to secondary region using LRS.
- Offers 16 nines of durability.



Geo-Zone-Redundant Storage (GZRS):

- Combines redundancy across availability zones with geo-replication.
- Copies data across three Azure availability zones in primary region and to secondary geographic region using LRS.
- Designed for maximum consistency, durability, and availability.



Read Access to Secondary Region:

RA-GRS and RA-GZRS:

- Enable read access to secondary region.
- Data available for read even when primary region is operational.
- Consider RPO for up-to-date data in secondary region.

Azure Storage Services

Azure Blobs:

Description:

- Object storage solution for text and binary data.

Features:

- Massively scalable.
- Unstructured data storage.
- Supports big data analytics via Data Lake Storage Gen2.

Ideal Use Cases:

- Storing massive amounts of data.
- Handling thousands of simultaneous uploads.
- Streaming video and audio.
- Backup, disaster recovery, and archiving.
- Storing data for analysis by on-premises or Azure-hosted services.

Azure Files:

Description:

- Managed file shares for cloud or on-premises deployments.

Features:

- Accessible via SMB or NFS protocols.
- Concurrently mountable by cloud or on-premises deployments.
- Supports caching on Windows Servers with Azure File Sync.

Key Benefits:

- Shared access without application compatibility concerns.
- Fully managed without hardware or OS management.
- Scripting and tooling support for easy administration.
- Resilient and always available.

Azure Queues:

Description:

- Service for storing large numbers of messages.

Features:

- Accessible worldwide via HTTP/HTTPS calls.
- Each message can be up to 64 KB in size.
- Commonly used for asynchronous processing and creating work backlogs.
- Integration with Azure Functions for triggering actions based on message receipt.

Azure Disks:**Description:**

- Block-level storage volumes for Azure VMs.

Features:

- Managed by Azure for resiliency and availability.
- Provisioning only required; Azure handles the rest.
- Virtualized disks offering greater reliability than physical disks.

Azure Tables:**Description:**

- Store large amounts of structured data.

Features:

- NoSQL datastore accepting authenticated calls from within and outside Azure.
- Ideal for structured, non-relational data storage.
- Enables building hybrid or multi-cloud solutions with data always available.

Azure Data Migration Options

Azure Migrate:**Description:**

Helps migrate from on-premises environments to Azure.

Features:

- Unified migration platform.
- Range of assessment and migration tools.
- Integrates with other Azure services.

Azure Data Box**Description:**

Physical migration service for transferring large data amounts.

Features:

- Accelerated data transfer with proprietary Data Box storage device.
- Secure transfer via regional carrier.

- Ideal for data sizes larger than 40 TBs with limited network connectivity

Azure Migrate Tools:

- **Azure Migrate: Discovery and Assessment:**
 - Identifies on-premises servers for migration.
- **Azure Migrate: Server Migration:**
 - Migrates VMware, Hyper-V, physical servers, and more to Azure.
- **Data Migration Assistant:**
 - Assesses SQL Servers for migration readiness.
- **Azure Database Migration Service:**
 - Migrates on-premises databases to Azure VMs running SQL Server, Azure SQL Database, or SQL Managed Instances.
- **Azure App Service Migration Assistant:**
 - Assesses on-premises websites for migration to Azure App Service.

Azure Data Box Use Cases:

Import Scenarios:

- Onetime migration from on-premises to Azure.
- Moving media libraries from offline tapes to Azure.
- Migrating VM farms, SQL servers, and applications to Azure.
- Moving historical data for analysis and reporting using HDInsight.

Export Scenarios:

- Disaster recovery by restoring Azure data to on-premises network.
- Meeting security requirements by exporting data out of Azure.
- Migrating back to on-premises or another cloud service provider.

Azure File Movement Options:

AzCopy

Description:

Command-line utility for copying blobs or files to/from Azure storage accounts.

Features:

- Upload, download, copy, and synchronize files.
- Works with other cloud providers.
- Supports one-directional synchronization.

Azure Storage Explorer:

Description:

Standalone app for graphical management of files and blobs in Azure Storage Account.

Features:

- Works on Windows, macOS, and Linux.
- Uses AzCopy for file and blob management tasks.
- Upload, download, and move files between storage accounts.

Azure File Sync:

Description:

Tool to centralize file shares in Azure Files while maintaining compatibility with Windows file servers.

Features:

- Bi-directional sync between local Windows server and Azure.
- Access data locally using SMB, NFS, or FTPS.
- Scalable caching across the world.
- Replacement of failed local servers.
- Cloud tiering for efficient storage management.

Azure Directory Services:

Microsoft Entra ID:

- Cloud-based identity and access management service.
- Provides sign-in access to Microsoft and custom cloud applications.
- Integrates with on-premises Active Directory for identity management.
- Detects suspicious sign-in attempts for enhanced security.

Users of Microsoft Entra ID:

- IT administrators for access control.
- App developers for adding functionality to applications.
- Users for self-service maintenance actions.
- Online service subscribers like Microsoft 365 and Azure users

Functionalities of Microsoft Entra ID:

- Authentication: Verify identity, password reset, multifactor authentication.
- Single sign-on: Simplify access to multiple applications.
- Application management: Manage cloud and on-premises apps.
- Device management: Register and manage devices for access control.

Integration with On-Premises AD:

- Connect with Microsoft Entra Connect for synchronized user identities.
- Enables consistent identity experience between cloud and on-premises environments.

Microsoft Entra Domain Services:

- Managed domain services in Azure.
- Provides domain join, group policy, LDAP, and Kerberos/NTLM authentication.
- Allows running legacy applications in the cloud without managing domain controllers.

- Integrates with existing Microsoft Entra tenant for seamless access and resource security.

Operation of Microsoft Entra Domain Services:

- Deployment of domain controllers in Azure regions as a replica set.
- Managed by Azure platform, including backups and encryption.
- One-way synchronization from Microsoft Entra ID to Microsoft Entra Domain Services.
- Identity information synchronized with Microsoft Entra ID through Microsoft Entra Connect in hybrid environments.

Azure Authentication Methods:

Authentication Overview

- Process of establishing identity for users, services, or devices.
- Involves providing credentials to prove identity.
- Azure supports various authentication methods for enhanced security and convenience.

Single Sign-On (SSO)

- Allows users to sign in once and access multiple resources.
- Simplifies user experience by reducing the number of passwords to remember.
- Requires trust between applications and providers for seamless access.

Multifactor Authentication (MFA)

- Prompts users for an extra form of identification during sign-in.
- Enhances security by requiring two or more authentication elements.
- Elements include something the user knows, has, or is, such as a PIN, phone code, or biometric property.

Microsoft Entra Multifactor Authentication

- Microsoft service providing multifactor authentication capabilities.
- Enables users to choose additional authentication forms during sign-in.

Passwordless Authentication:

- Removes the need for passwords, offering enhanced convenience.
- Users authenticate with something they have, are, or know, without entering passwords.
- Offers options like Windows Hello for Business, Microsoft Authenticator app, and FIDO2 security keys.

Passwordless Authentication Options:

- Windows Hello for Business:
- Ideal for individual Windows PC users.
- Biometric and PIN credentials tied to user's PC for enhanced security.

Microsoft Authenticator App:

- Turns smartphones into strong, passwordless credentials.

- Users confirm sign-in via phone notification, matching displayed numbers, and biometric/PIN.

FIDO2 Security Keys:

- Unphishable passwordless authentication method.
- Users register security keys for authentication, increasing account security.
- Keys come in various form factors like USB, Bluetooth, or NFC.

Azure External Identities:

Definition:

- External identity means individuals or entities outside your organization.
- Microsoft Entra External ID allows secure interaction with these external users.

Capabilities:

- Business to Business (B2B) Collaboration:
- Work with external users using their own logins.
- These users are seen as guests in your system.

B2B Direct Connect:

- Establish trust with another Microsoft Entra organization.
- Useful for shared collaboration, like Teams channels.

Microsoft Azure Active Directory Business to Customer (B2C):

- Publish apps for consumers.
- Manage user logins using Azure AD B2C.

Usage

- Collaborate with external partners or vendors easily.
- Invite external users or social identities to your system.
- Review and manage access for external users efficiently.

Azure Conditional Access:

Definition:

- Conditional Access is a tool in Microsoft Entra ID that decides who can access resources based on identity signals.
- Signals include user identity, location, and device used.

Purpose:

- Empower users to work from anywhere while safeguarding organization assets.
- Provide a more secure login experience, like asking for extra verification if needed.

How It Works:

- Collects signals during sign-in.
- Makes a decision based on these signals.
- Enforces the decision by allowing or denying access, or asking for extra verification.

Example:

- If a user signs in from their usual place, they might get full access.
- If it's an unusual location, access might be blocked or require extra verification.

Usage:

- Require extra verification (like MFA) for certain roles or locations.
- Allow access only from approved applications or managed devices.

Azure Role-Based Access Control (Azure RBAC):

Purpose

- Manage who can access what in your Azure cloud environment.
- Ensure that users have the necessary access to perform their tasks without granting excessive permissions.

How It Works:

- Azure provides predefined roles with specific responsibilities, such as "Owner," "Contributor," or "Reader."
- Custom roles can also be created with tailored permissions to suit unique requirements.
- Users or groups are assigned to these roles, inheriting the associated permissions.

Application to Resources:

- Roles are applied to different scopes within Azure, including management groups, subscriptions, resource groups, or individual resources.
- Permissions set at a higher level, such as a subscription, are inherited by all resources within that scope.

Enforcement:

- Azure RBAC governs access to Azure resources whenever actions are initiated through Azure Resource Manager.
- Resource Manager organizes and secures resources, ensuring that users have the appropriate permissions when interacting with resources.

Example:

- An "Owner" role might be assigned to a user responsible for managing all resources within a subscription, granting full control.
- A "Reader" role could be assigned to a user who only needs to view resources without making any changes.

Key Concept:

- Azure RBAC follows an "allow" model, meaning that users are allowed to perform actions within the scope of their assigned roles.
- If a user has multiple role assignments granting conflicting permissions, they will have the combined permissions of all assigned roles.

Zero Trust Model:

Definition:

A security approach that always double-checks requests as if they're coming from an unsafe place.

Guiding Principles:

- **Verify Everyone:**
 - Always check who's asking for access.
- **Use Minimal Access:**
 - Give only the necessary permissions.
- **Expect the Worst:**
 - Assume there might be a breach and prepare accordingly.

Adaptation:

Instead of trusting devices within the network, it now verifies every user's identity no matter where they are.

Defense-in-Depth

Objective:

Defense-in-depth is like building a series of walls around your data to keep it safe from intruders.

Strategy:

Assume attackers will try to breach your defenses and create multiple layers of protection to slow them down and minimize damage.

Layers of Defense

Physical Security:

Lock doors, install cameras, and use biometric scanners to keep intruders out of your physical data centers.

Identity & Access:

Control who can access what by managing user accounts, passwords, and permissions carefully.

Perimeter:

Block malicious traffic before it enters your network with firewalls and other security tools.

Network:

Control the flow of data between devices and networks to prevent unauthorized access.

Compute:

Keep your computers and servers secure by installing antivirus software and applying security updates regularly.

Application:

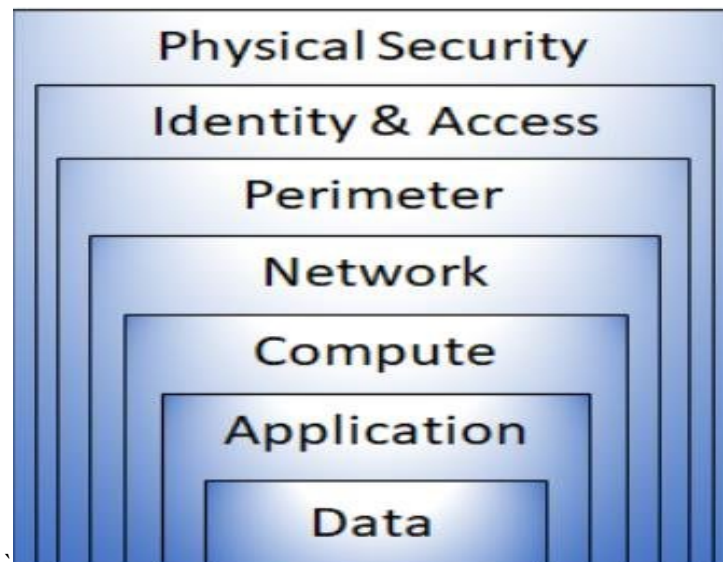
Make sure your software is free from bugs and vulnerabilities to prevent attackers from exploiting weaknesses.

Data:

Protect your sensitive information by encrypting it and controlling who can access it.

Azure's Role:

Azure provides tools and services to help you implement defense-in-depth, such as Azure Active Directory for managing user identities and Azure Security Center for monitoring and managing security across your Azure environment.

**Microsoft Defender for Cloud:**

Defender for Cloud is a security monitoring tool that keeps your cloud, on-premises, hybrid, and multi-cloud environments safe from cyber threats.

Key Features:

- Monitors and protects your resources.
- Provides guidance and notifications to improve security posture.
- Easy deployment as it's natively integrated with Azure.

Protection Everywhere:

- Automatically monitors Azure services, but can extend protection to on-premises and other cloud environments.
- Can deploy agents to gather security data from non-Azure machines.
- Uses Azure Arc to extend protection to hybrid and multi-cloud environments.

Azure-Native Protections:

- Detects threats targeting Azure services like Azure App Service and Azure SQL.
- Classifies data in Azure SQL and provides vulnerability assessments for Azure SQL and Storage services.
- Helps in network hardening by limiting access to virtual machine ports.

Hybrid Resources:

- Extends protection to non-Azure servers in hybrid cloud environments.
- Customized threat intelligence and prioritized alerts for hybrid environments.

Other Clouds:

- Can protect resources in other clouds like AWS and GCP.
- Offers various protections including CSPM features and threat detection for AWS resources.

Assess, Secure, and Defend:

- Continuously assesses your security posture and identifies vulnerabilities.
- Helps in securing resources with Azure Security Benchmark and secure configuration standards.
- Provides security alerts and advanced threat protection features to defend your environment.

Security Alerts:

- Generates alerts when threats are detected, providing details and remediation steps.
- Supports fusion kill-chain analysis to correlate alerts and understand attack campaigns.

Advanced Threat Protection:

- Provides advanced threat protection for virtual machines, SQL databases, containers, web applications, and networks.
- Includes features like just-in-time access and adaptive application controls for enhanced security.

LP 3

Azure Cost Factors

Resource Type:

- Different types of resources have different costs based on settings and region.
- Example: Storage account settings and virtual machine sizes affect costs.

Consumption:

- Pay-as-you-go model charges for resources used during a billing cycle.
- Reserved resources offer discounts for committing to usage in advance.

Maintenance:

- Unused resources should be deprovisioned to control costs.
- Resource groups help organize and manage resources efficiently.

Geography:

- Azure resources deployed in different regions may have varying costs.
- Costs are influenced by factors like power, labor, taxes, and fees.

Network Traffic:

- Data transfers within Azure datacenters are free.
- Outbound data transfer costs vary based on billing zones.

Subscription Type:

- Different subscription types offer varying usage allowances and free products.
- Example: Azure free trial provides access to certain products for 12 months.

Azure Marketplace:

- Offers third-party solutions and services with varying billing structures.
- Costs may include both Azure services and vendor services/expertise.
- All solutions comply with Azure policies and standards.

Pricing and Total Cost of Ownership (TCO) Calculators

Pricing Calculator:

- Estimates the cost for provisioning resources in Azure.
- Helps you understand the cost of individual resources or an entire solution.
- Focuses on the cost of provisioned resources like compute, storage, and network.
- No services are provisioned, and no charges are incurred when using the calculator.

TCO Calculator:

- Helps compare costs of running on-premises infrastructure versus Azure Cloud infrastructure.
- Requires input of current infrastructure configuration, including servers, databases, storage, etc.
- Compares anticipated costs for current environment with Azure environment supporting the same requirements.
- Includes assumptions like power and IT labor costs to provide a comparison of costs.

Microsoft Cost Management Tool

- Helps monitor and manage Azure resource costs
- Provides insights to avoid unexpected expenses

What is Cost Management?

- Quickly check Azure resource costs
- Create alerts and budgets for cost control

Cost Analysis

- Visualizes Azure costs by various parameters
- Analyzes costs by billing cycle, region, resource, etc.

Cost Alerts

- **Budget alerts:** Notify when spending exceeds defined limits
- **Credit alerts:** Notify when Azure credit commitments are consumed
- **Department spending quota alerts:** Notify when department spending reaches fixed thresholds

Budgets

- Set spending limits for Azure
- Define budgets based on subscription, resource group, etc.
- Trigger alerts and automate actions based on budget conditions

What are Resource Tags

Resource tags are labels that you can assign to your Azure resources. They help you categorize and organize your resources.

Benefits of Using Tags

- **Manage Resources Easily:** Tags help you find and organize related resources, such as by application, workload, environment or other criteria.
- **Track Costs:** Tags let you group resources by category so you can monitor costs by department, workload, project etc. This helps you allocate costs and make budgets.
- **SLA Management:** Tags help you classify resources by criticality, so you can define appropriate SLAs (uptime guarantees).

- **Security Classification:** Tags enable you to categorize data by security level. For example "Confidential", "Internal", "Public".
- **Meet Compliance Requirements:** Tags help you identify resources aligned with industry/regulatory standards like ISO or SOX.
- **Automation:** Tags enable you to group resources and perform automated tasks like updates, monitoring, scaling using tools like Azure DevOps.

How to Manage Tags

- Portal, CLI, PowerShell, API, ARM Templates can all be used to add, edit or remove tags
- Azure Policy helps set tagging rules, ensure standards
- Tags defined at one level (resource, RG, subscription) don't flow to other levels automatically

Tagging Best Practices

- Name-Value format. Example: Environment-Production, AppName-CRM, OwnedBy-John
- Not mandatory to have same tags on all resources. For example, "Criticality" tag may only be needed for mission-critical resources.
- Reusing common tags across resources, resource groups and subscriptions improves management.

Microsoft Purview Purpose

Data Governance, Risk, and Compliance:

- Provides a unified view of data across on-premises, multicloud, and software-as-a-service platforms.
- Enables staying up-to-date on the data landscape through automated discovery, sensitive data classification, and end-to-end data lineage.

Two Main Solution Areas:

Risk and Compliance:

- Utilizes Microsoft 365 features like Teams, OneDrive, and Exchange for managing and monitoring data.
- Helps protect sensitive data, identify risks, and ensure regulatory compliance across clouds, apps, and devices.

Unified Data Governance:

- Manages on-premises, multicloud, and software-as-a-service data.
- Capabilities extend to Azure, SQL, Hive databases, and other clouds like Amazon S3.
- Enables creating a comprehensive map of the data estate with classification and lineage, identifying sensitive data locations.
- Facilitates secure data consumption, insights generation, and scalable access management.

Azure Policy Purpose

Service in Azure for setting rules to manage resources and ensure they meet standards.

Defining Policies:

- Allows creating individual rules or groups of rules (initiatives).
- Checks resources for compliance and can stop non-compliant resources from being made.

Policy Implementation:

- Rules can be applied to specific resources, groups, or subscriptions.
- If a rule is set at a high level, it applies to everything below it automatically.

Built-in Policy Definitions:

- Ready-made rules for common services like Storage, Networking, and Security Center.
- Can check both existing and new resources.

Automatic Fixes:

- Can automatically correct issues with resources that don't meet rules.
- Can also exempt specific resources from automatic fixes.

Integration with Azure DevOps:

- Works with DevOps pipelines for managing application deployment policies.

Azure Policy Initiatives:

- Bundles related rules together to track compliance for bigger goals.

Example of Initiative:

- "Enable Monitoring in Azure Security Center" makes sure all resources are monitored for security.
- Includes rules like checking for unencrypted databases and missing security measures.

Resource Locks Purpose

- Prevent accidental deletion or changes to resources in Azure.
- Azure RBAC policies might not completely prevent users from mistakenly deleting critical resources.
- Resource locks provide an additional layer of protection against such mishaps.
- They can be applied to individual resources, resource groups, or entire subscriptions.
- Resource locks are inherited, meaning they affect all resources within a group if applied to the group.

Types of Resource Locks:

- **Delete Lock:** Stops users from deleting a resource, but they can still read and modify it.
- **ReadOnly Lock:** Allows reading a resource but prevents deletion or updates, like the Reader role.

Managing Resource Locks:

- Resource locks can be managed through various Azure tools like the portal, PowerShell, or Azure CLI.
- In the Azure portal, go to the Settings section of any resource's Settings pane to view, add, or delete locks.

Handling Locked Resources:

- To make changes to a locked resource, the lock must be removed first.
- Resource locks take precedence over RBAC permissions, so even owners must remove locks before performing restricted actions.

Service Trust Portal Purpose

- The Service Trust Portal is a platform that offers information and tools regarding Microsoft's security, privacy, and compliance practices.
- It provides details about how Microsoft safeguards its cloud services and the data of its customers.
- Users need to sign in with their Microsoft cloud services account to access certain resources.
- Acceptance of the Microsoft non-disclosure agreement may be required to view compliance materials.

Accessing the Portal:

The portal can be accessed at <https://servicetrust.microsoft.com/>.

Features and Content:

The main menu provides access to different categories:

- Service Trust Portal: Returns to the home page.
- My Library: Allows users to save documents for quick access and set up notifications for updates.
- All Documents: Provides a central location for all documents on the portal, from where users can pin documents to their personal library.

Tools for Interacting with Azure

Azure Portal:

- Web-based console for managing Azure subscriptions.
- Allows building, managing, and monitoring cloud resources using a graphical interface.
- Customizable dashboards and accessibility options available.

Azure Cloud Shell:

- Browser-based shell tool within the Azure portal.
- Supports Azure PowerShell and Azure CLI.
- Authentication to Azure credentials, and support for both Azure PowerShell and Azure CLI.

Azure PowerShell:

- Shell for running commands (cmdlets) to manage Azure resources.
- Calls Azure REST API for management tasks.
- Used for setting up, tearing down, and maintaining resources, as well as infrastructure deployment.
- Available on Windows, Linux, and Mac platforms.

Azure Command Line Interface (CLI):

- Equivalent to Azure PowerShell but uses Bash commands.
- Offers similar capabilities for managing Azure resources.
- Installable on Windows, Linux, and Mac platforms.
- Accessible through Azure Cloud Shell.

Summary

- Azure Portal is a user-friendly interface for managing Azure resources visually.
- Azure Cloud Shell is a convenient way to access PowerShell or CLI directly within your browser, without any installation.
- PowerShell and CLI are command-line tools for scripting and automation, available on various operating systems.

Azure Arc

Purpose:

Simplify management of hybrid and multi-cloud environments.

Key Points:

- Extends Azure's capabilities to on-premises and multi-cloud setups.
- Provides a unified management platform using Azure Resource Manager (ARM).
- Allows managing non-Azure resources as if they were in Azure.
- Offers consistency in managing virtual machines, Kubernetes clusters, and databases.
- Utilizes familiar Azure services and management tools.
- Supports both traditional IT operations and DevOps practices.
- Enables custom configurations for better abstraction and management.

Capabilities Outside Azure:

Manages servers, Kubernetes clusters, Azure data services, SQL Server, and virtual machines hosted outside Azure (virtual machines feature is in preview).

Azure Resource Manager (ARM) and ARM Templates

Azure Resource Manager (ARM):

- Manages deploying and managing resources in Azure.
- Acts as a central hub for all operations on Azure resources.
- Ensures consistent results across different Azure tools.

Benefits of ARM:

- **Infrastructure Management:**
 - Manages infrastructure through templates rather than manual scripts.
 - Handles resources as a group, streamlining management.
 - Supports consistent redeployment throughout development phases.
 - Defines dependencies between resources for correct deployment order.
- **Access Control and Billing:**
 - Integrates Role-Based Access Control (RBAC) for security.
 - Organizes resources using tags for easier billing.

ARM Templates:

- Declarative JSON files describing resources for deployment.
- Ensures correct resource creation and connection.
- Orchestrates parallel creation of multiple instances.
- Executes scripts for pre- or post-deployment configurations.

Benefits of ARM Templates:

- **Declarative Syntax:**
 - Simplifies deployment by declaring desired state without programming.
 - Offers repeatable deployments ensuring consistency.
 - Orchestrates deployment operations, including parallel creation.
- **Modular and Extensible:**
 - Breaks templates into reusable components for easy management.
 - Incorporates PowerShell or Bash scripts for enhanced functionality.

Bicep:

- Declarative language for deploying Azure resources.
- Defines infrastructure and configurations.
- Simplifies syntax compared to JSON templates.

Benefits of Bicep:

- **Comprehensive Support:**
 - Immediately supports all Azure services without waiting for updates.
 - Requires no programming knowledge for syntax understanding.
- **Consistent Deployments:**
 - Ensures consistent deployment across different environments.
 - Orchestrates resource creation in correct order for efficient deployment.
- **Modular Design:** Breaks code into reusable modules for simplified development.

Azure Advisor

What is Azure Advisor?

- It evaluates your Azure resources.
- Provides recommendations to enhance reliability, security, performance, operational excellence, and cost-efficiency.

Purpose:

- Helps save time on cloud optimization.
- Offers actionable suggestions for immediate action, postponement, or dismissal.

Accessing Recommendations:

- Available through Azure portal and API.
- Notifications can be set up for new recommendations.

Advisor Dashboard:

- Personalized recommendations for all subscriptions.
- Filters available for specific subscriptions, resource groups, or services.

Recommendation Categories:

- **Reliability:** Ensures continuity of critical applications.
- **Security:** Identifies threats and vulnerabilities.
- **Performance:** Enhances application speed.
- **Operational Excellence:** Improves process efficiency and deployment practices.
- **Cost:** Optimizes Azure spending.

Visual Representation:

- Dashboard displays recommendations categorized by type.
- Helps users easily identify areas for improvement.

Overall Purpose:

Streamlines cloud management by providing actionable insights for better resource utilization and cost savings.

Azure Service Health Notes:

Purpose: Helps you track the health of your Azure resources and infrastructure.

Components:

- **Azure Status:** Shows the overall health of Azure globally, highlighting service outages.
- **Service Health:** Gives updates on specific Azure services and regions you use, including outages and planned maintenance.
- **Resource Health:** Provides details about the health of your individual Azure resources, like virtual machines.

Functionality:

- Accessible through the Azure portal.
- Alerts can be set up to notify you about service issues or changes affecting your resources.

Benefits:

- Provides a complete view of your Azure environment, from global status to specific resources.
- Historical alerts are saved for future review and analysis.

Support: Offers links to support in case your workload is affected by an event.

Azure Monitor

What is Azure Monitor?

- Platform for collecting, analyzing, visualizing, and acting on data from various sources.
- Monitors Azure, on-premises, and multi-cloud resources.

Components:

- **Data Collection:** Gathers logging and metric data from different layers of your architecture.
- **Storage:** Stores data in central repositories.
- **Visualization:** Displays real-time and historical performance data across your architecture.
- **Alerting:** Notifies teams about critical events in real-time and triggers autoscaling.

Azure Log Analytics:

- Tool in the Azure portal for writing and running log queries.
- Supports simple and complex queries, statistical analysis, and data visualization.

Azure Monitor Alerts:

- Automated alerts triggered when thresholds are crossed.
- Alerts based on log events or metrics.
- Uses action groups to configure notifications and actions.

Application Insights:

- Monitors web applications running in Azure, on-premises, or other cloud environments.
- Tracks request rates, response times, failure rates, dependency rates, page views, load performance, user and session counts, and more.
- Can be configured using SDK or Application Insights agent for various programming languages.